

Re-wiring Activity of Malicious Networks

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Abstract. This paper studies the AS-level *re-wiring dynamics* (changes in the connectivity) of malicious networks. Anecdotal evidence suggests that some malicious ASes that are primarily involved in nefarious activities on the Internet, were sequentially de-peered by providers before their final cut-off (as occurred in the well-publicized cases of Atrivo/InterCage). We present the first systematic study of the re-wiring dynamics of malicious ASes. We tracked the ASes that were listed by Hostexploit over the last two years and compared their AS-level re-wiring dynamics with non-reported ASes. Using a publicly available dataset of Customer-Provider (CP) relations in the Internet's AS graph, we studied how interconnection between autonomous systems evolves, both for ASes that provide connectivity for attackers and ASes that were not reported as malicious. We find that malicious networks are more aggressive both in forming links with providers and changing their upstream connectivity than other ASes. Our results indicate that the re-wiring dynamics of the networks that host attacks are stable over time, despite the evolving nature of the attacks themselves, which suggests that existing defense mechanisms could benefit from incorporating these features.

1 Introduction

Securing the Internet's routing system has been a concern for both network operators and protocol designers for nearly fifteen years. One of the frequently stated reasons for securing the Internet's interdomain routing protocol, the Border Gateway Protocol (BGP), is that attackers may use BGP to launch their attacks and hide their traces. Previous work has exposed some specific techniques that attackers use; for example, Ramachandran *et al.* observed that some attackers send spam from short-lived prefixes [13]. Nevertheless, little is known about how ASes that host these attackers exploit BGP to provide them with further protection. There have been some publicized cases of malicious ASes—major hubs of illegal activity—namely Atrivo/InterCage and VolgaHost, that were observed to get frequently de-peered. Eventually they were officially reported and cut off by all providers in September 2008 and January 2011, respectively. This practice of frequent change in upstream connectivity and eventual cut-off may constitute a feature that potentially characterizes malicious ASes, regardless of the type of attacks they launch which varies across time. We refer to the activity that is related with change of connectivity of an AS as *re-wiring activity* of this AS.

This paper presents the first systematic study of the re-wiring activity of malicious ASes, with the goal of improving our understanding of how malicious networks exploit interconnection through different upstream ASes to cover their tracks. Rather than attempting to detect any individual type of attack (e.g., spam, denial of service), we

characterize the re-wiring activity of malicious networks that are primarily responsible for attacker activities. We identify features of the re-wiring behavior that may be more stable across time than the characteristics of any single attack. We believe that ultimately certain aspects of routing behavior may serve as invariants for detecting malicious infrastructure, even as the attacks themselves evolve.

We draw the following conclusions from our study:

- *Malicious Enterprise Customers (ECs) on average change their upstream connectivity more aggressively than non reported ECs.* We offer a new class of observations on the AS-level re-wiring activity of malicious ECs. ECs are typically stub networks (Section 3). We find that, on average, malicious ECs change their upstream connectivity more frequently and link with a larger set of providers throughout their lifetime than non-reported ECs. We observe that malicious ECs link on average with a total of twelve providers from 1998–2010, whereas the rest of ECs link with only about four providers on average. Also, malicious ECs’ peering sessions last for less time; the top 5% of CP links formed by malicious ECs are observed for 14–31 consecutive snapshots of the AS graph, whereas the top 5% of the CP links formed by the rest of ECs are observed for 27–51 consecutive snapshots.
- *Malicious ECs prefer to attach to popular providers as non reported ECs do when they first appear, but they attach to less popular providers when they re-wire later in their lifetime.* In 2009–2010, all malicious ECs that were observed for the first time used the *most-preferred providers* (the providers that are responsible for the most CP links generated by non-reported ECs at that snapshot) for transit, as the rest of the ECs did (Section 4). During the same time period, the malicious ECs that were observed to re-wire, attached to providers from which only a fraction of 30–50% were among the top providers preferred by non-reported ECs.

Our results highlight some findings that have implications for the design of more effective defense mechanisms against Internet attacks, which we discuss in more detail in Section 5. Even though an AS may be acting maliciously, network administrators in other networks may not be aware of it. Instead, if AS rewiring activity is monitored and potential malfeasance is reported, then this information may be useful to network administrators. For example, in the case of AS-Atrivo, even if not all network administrators knew that AS-Atrivo was acting maliciously or the details of the attacks, it may have been helpful to know that its re-wiring activity was suspicious, in case they receive suspicious traffic originating from that AS or if AS-Atrivo tries to connect directly with them. The re-wiring features are independent of the evolving nature of the attacks and also more difficult for the attackers to tamper, incorporating them may provide significant gains over existing defense mechanisms or complement them.

The rest of this paper is organized as follows. In Section 2, we give an overview of related work. In Section 3, we describe the datasets we used in our analysis. Section 4 presents our findings and Section 5 offers some observations about the implications of our findings with regard to existing defense mechanisms. We conclude in Section 6.

2 Related Work

Previous studies have investigated various types of malicious activity in isolation such as worms, botnets, spam, and Internet scam infrastructure [2, 6, 7, 9–12, 15, 18] to propose detection solutions. Kim *et al.* [9] designed a system that generates worm signatures based on the analysis of the prevalence of portions of flow payloads. Xie *et al.* [19] analyzed a three-month sample of emails to characterize spamming botnets, properties of botnet IP addresses, sending patterns and developed a system called AutoRE that generated regular expression signatures for detection of botnet-based spam emails and botnet membership. Anderson *et al.* [1] developed a measurement system called Spamsscatter that mines emails in real-time, follows the link embedded at the email, and clusters the destination Web sites using image shingling to capture graphical similarity between rendered sites. In contrast, we focus on networks whose primary purpose is to instrument these activities, and we attempt to understand their re-wiring behavior at the AS level.

There exists a limited number of studies that have investigated spammers' use of the routing infrastructure to facilitate their activities [16, 17]. These studies have provided evidence of the fact that spammers advertise routes to hijacked IP prefixes for short amounts of time. Ramachandran *et al.* [13, 14] performed a large-scale study on the network-level properties of spam and quantified the extent to which spammers use this technique of short-lived BGP announcements to send spam. Also, Feamster *et al.* [4, 5] studied route advertisements for private or unassigned ("bogon") IP addresses. Our work investigates the re-wiring activity of networks that have been reported as malicious, and we compare them with non-reported networks.

3 Data

Our primary dataset consists of a list of networks that are reported as top in Internet criminal activities, in a period of three years (2009–2011) according to Hostexploit. There are a total of 129 distinct ASes. We hypothesize that malicious ASes have different business functions and incentives than most ASes, which may be reflected on the links they form with other ASes in the Internet. To understand the re-wiring behavior of these networks we obtain a publicly available dataset of customer-provider links formed among ASes over ten years (available here [3]), over which we track them. We consider ASes that have not ever been reported by Hostexploit to be legitimate.

Hostexploit reports. To study the behavior of malicious networks that are mostly engaged in Internet nefarious activities, we collected the list of ASes reported as most malicious networks by Hostexploit in global Internet activity reports issued in quarters of 2009–2011. Hostexploit is an open organization of international volunteers who are Internet professionals within the areas of web hosting, server management, DNS, Internet security, and intrusion detection systems, with a focus on creating awareness of cyber-crime activity. Hostexploit integrates and correlates data from multiple sources such as spam, malware, malicious URLs, spam bots, botnet command and control servers, phishing servers, exploit servers, and cyber-warfare intelligence provided by industry

AS Number	Name	Country
AS41947	WEBALTA-AS OAO Webalta	RU 
AS29073	ECATEL-AS AS29073, Ecatel Network	NL 
AS16138	INTERIAPL INTERIA.PL Autonomous System	PL 
AS10297	ENET-2 - eNET Inc.	US 

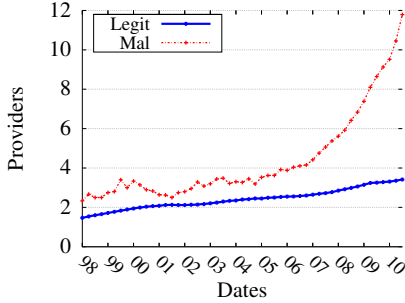
Fig. 1. A quarterly report from Hostexploit from the 1st quarter of 2011. The report ranks ASes by an index (HE) based on the activity of the AS, weighted by the size of its allocated address space.

partners. To provide a list of the worst networks, they rate each AS with a Hostexploit Index (HE), based on the activity of the AS weighted by the size of its allocated address space. HE is a proportional rather than an absolute index of AS maliciousness. Hostexploit has been issuing reports on top networks by their HE index periodically since 2009.

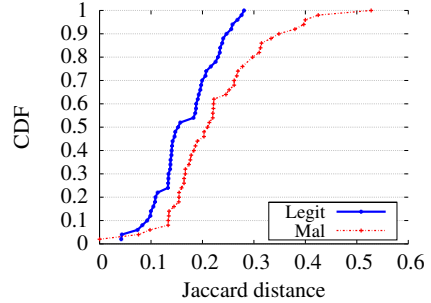
Figure 1 shows an example Hostexploit report for the first quarter of 2011. The report is accompanied by an analysis of the quarterly results. For example, they compare the new ranking results with the previous quarterly ranking, show the top worst networks by sector (spam, botnet hosting, etc.), show an analysis by country, etc.

Historic AS relationships. To understand the re-wiring behavior of malicious networks, we tracked these ASes' rewiring activity over a ten years. Dhamdhere *et al.* [3] collected BGP AS paths from BGP table dumps obtained from repositories at RouteViews and RIPE. They collected snapshots of AS paths; a snapshot refers to a period of 21 days (not an instant). During each snapshot, they collected at five different times the unique AS paths from all active monitors. Then, to obtain the primary Internet links (used most of the time) and to filter out the backup links (used during failures or overload conditions), they kept only the AS paths that appeared in the majority of the samples and ignored the rest (majority filtering algorithm). Then, they used the AS paths in each snapshot (those that had passed the majority filtering process) to infer the underlying AS topology and the relationships between adjacent ASes. Finally, they used the well-known algorithm proposed by Gao [8]. Gao's algorithm resulted in four types of AS relationships: Customer-Provider (CP), peering, sibling, and unknown. To understand the evolution of the Internet ecosystem they classified ASes into types depending on their function and business type, based on their observable topological properties. They classified ASes into Enterprise Customers (ECs), small transit provider, large transit provider, content access and hosting provider.

In this study, we focus only on the CP links of ECs (62 in total). We note that based on the method that the classes were derived, ASes of the same class are close in terms of the number of customers and peers they link with at each snapshot. As far as the number of providers they link with at each snapshot there is larger variability among



(a) Accumulation of providers.



(b) Distribution of the average Jaccard distance between two consecutive snapshots of links with providers.

Fig. 2. Malicious ECs on average attach to more providers and change their providers more frequently than legitimate ECs

the ASes, but our analysis is not relying on this metric. Instead, we are looking at the accumulation of distinct providers across time, the CP link duration and other features (see Section 4).

Dataset limitations. Our data has the following limitations: (1) The Hostexploit data may be inaccurate. (2) The methods that were used to infer the AS graph [3] have been shown to be inadequate. Most ASes are detected but a significant fraction of peering and backup links at the edges of the Internet are missed. (3) Dhamdhare *et al.*'s AS classification algorithm may introduce errors.

4 Results

We hypothesize that malicious ASes have different business functions and incentives than the rest of the ASes, and that these differences may be reflected on their re-wiring activity. For example, we suspect that malicious ASes may get frequently de-peered from their providers and that they need to re-wire more frequently than legitimate ASes. To test our hypothesis, we survey the wiring trends of malicious networks; (1) re-wiring frequency and (2) attachment preference. To determine whether our findings may be suitable for characterizing network re-wiring activity, we compare the behavior of malicious ASes to that of non-reported ASes over a ten-year period. We present the results of our analysis for Enterprise Customers (ECs) only.

4.1 Re-wiring Frequency

To understand the re-wiring dynamics of malicious networks and, more specifically, how these networks change their upstream connectivity, we examine three features; (1) the number of providers they link with throughout their lifetime, (2) how frequently

they change providers and (3) how long their CP links last. To determine whether these features may be suitable for characterizing the re-wiring activity of malicious ECs, we compare the behavior of malicious ECs to that of the rest of the ECs over ten years.

Malicious ECs on average connect to more providers than legitimate ECs Do. To investigate how malicious ECs link with their providers we first look at the total number of providers they link with throughout their lifetime. To compute the cumulative number of providers that an EC has, on average, first we track each EC for the duration of its lifetime; we add the total number of providers it links with to the total number of providers for its group (malicious or rest of EC). Then, for every snapshot, we compute the average by dividing the cumulative total number of providers of the EC group with the number of ASes in the EC class. We note that across time the size of the group increases as new ECs appear, and as does the cumulative number of providers for the group of EC. The rate with which the two numbers increase does not stay the same across time, so some snapshots exhibit a small decrease in the average cumulative number of providers. Figure 2(a) shows the cumulative number of providers that ECs attach to, on average, for two cases: malicious ECs and the rest of ECs. On average, malicious ECs link with more providers throughout their lifetime than the rest of the ECs do.

Malicious ECs link on average with a total of twelve providers during the period 1998–2010, whereas the rest of ECs link on average with only about four providers. The top ECs by the total number of providers are AS 3, which connected to 83 distinct providers; AS 26415, which linked with a total of 68 providers; and the private AS 65000, linked to 20 different providers. On the other hand, the malicious EC that we observed to link with the most providers (a total of 83) through its lifetime was AS 23456. We note that this AS is reserved by IANA and is used for backward compatibility between old (2-byte AS support) and new (4-byte AS support) BGP speakers. Because it is reserved for backwards compatibility, this AS is likely not a single AS, but rather a group of ASes; we must investigate this further to better understand the individual ASes that are perpetrating suspicious activity.

Malicious ECs on average change upstream providers more frequently than legitimate ECs. To quantify the aggressiveness of an EC in changing its upstream connectivity, we consider the distance between the set of the AS's providers for two consecutive snapshots. We use the Jaccard distance as a metric of distance between the set of providers of two consecutive snapshots. For example, a Jaccard distance of 0.8 indicates that 80% of the links seen in the two snapshots are observed in one of the two snapshots but not in both. We calculated the Jaccard distance between two consecutive snapshots for each AS throughout its lifetime. Figure 2(b) shows the distribution of the Jaccard distance values for malicious ECs and the rest of the ECs throughout their lifetime. We observe that malicious ECs are more aggressive on average in changing their upstream connectivity than the rest of ECs.

The fact that malicious ECs change their upstream connectivity more frequently than legitimate ECs may be a reflection of the following: (1) malicious ECs are noticed by their providers regarding their attacker activities, they get de-peered and they are forced to change upstream connectivity more frequently than legitimate ECs, (2) malicious ECs attempt to find providers with less strict restrictions, which may make it easier for

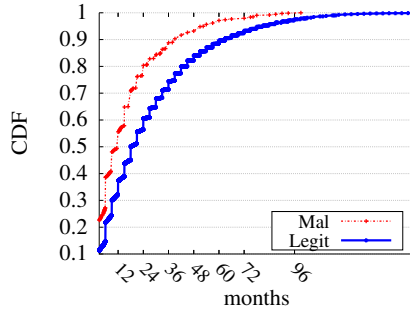


Fig. 3. Distribution of the average duration of CP links formed by ECs with their providers. Malicious ECs on average form shorter duration links than the rest of the ECs.

them to launch their attacks, (3) malicious ECs attempt to avoid accountability or legal consequences of their activities.

Malicious ECs on average form shorter duration CP links than legitimate ECs. To better understand the behavior of malicious ECs, we consider the duration of the links they form with their providers. To compute the CP link duration, we proceed as follows: First, for every snapshot, we determine the links that are present. Second, for each link, we determine whether it is present in the previous snapshot. Finally, for each link, we measure the total number of consecutive snapshots it was present. In cases where a link appears multiple times throughout 1998–2010 we consider the average duration of that link. Figure 3 shows the distribution of average CP link duration of malicious ECs and the rest of ECs. We observe that malicious ECs on average form CP links that are shorter in duration than the CP links that legitimate ECs form.

Figure 3 shows that approximately 22% of the CP links formed by malicious EC and about 10% of CP links formed by the rest of ECs were observed for only one snapshot. For example ASes in AS 23456, which linked with the largest number of providers, formed CP links with ASes 30083, 5617, and 7643, which appeared for only one snapshot. Examples of short-lived CP links formed by non-malicious ECs include AS 20195–AS 6395, AS 39709–AS 28870, and AS 3748–AS 4554. These links were observed for only one snapshot. On the other hand, as far as long-lived CP links, the top 5% of CP links formed by malicious ECs are observed for 14–31 consecutive snapshots, whereas the top 5% of the CP links formed by non-malicious ECs are observed for 27–51 consecutive snapshots. The most long-lived CP links by malicious ECs are links AS 14280–AS 6327 which was observed for 33 consecutive snapshots, AS 17974–AS 7713 for 31 consecutive snapshots and AS 13174–AS 1299 for 30 consecutive snapshots. The most long-lived CP links by legitimate ECs were observed for 51 consecutive snapshots (i.e., the entire period from 1998 to 2010); some examples are: AS 8581–AS 5408, AS 2508–AS 2907, AS 7065–AS 701, AS 10357–AS 7066, and AS 1104–AS 1103.

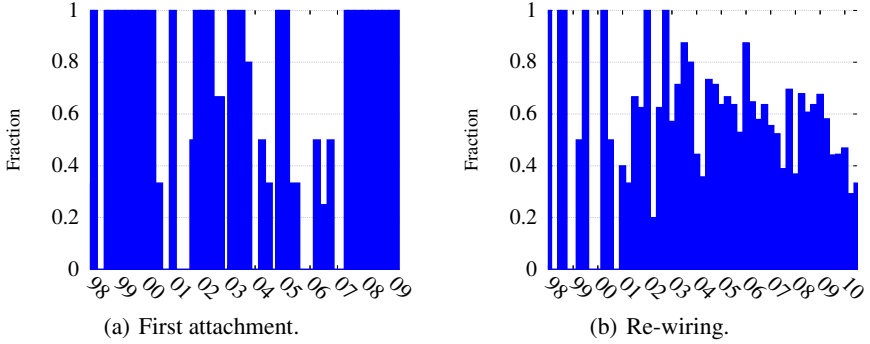


Fig. 4. Fraction of malicious ASes providers' that belong to the most preferred providers of all ECs for each snapshot. Malicious ECs in some cases attach to the most popular providers when they first appear, but not when they rewire. This is especially true for the more recent years.

4.2 Attachment Preference

In this section, we study the providers that malicious ECs connect to. Our goal is to determine whether malicious ECs have different attachment preferences than the rest of the ECs. We define the *most preferred providers* as the providers that are responsible for the most CP links generated by non-reported ECs at that snapshot. We find that malicious ECs do not connect to the most preferred providers as non-reported ECs do.

First, at each snapshot, we determine the most preferred providers as follows: we calculate the total number of CP links generated by non-reported ECs at each snapshot and we extract the providers that are responsible for at least 60% of the total at that snapshot. This set comprises the most preferred providers of non-reported ECs at each snapshot. Second, for each snapshot, we determine the providers that malicious ECs link with and whether those links are first-time attachments or re-wirings. Finally, at each snapshot, we calculate the fraction of the providers that malicious ECs link with that also belong to the most preferred providers of non-reported ECs. We calculate this fraction separately for the cases of first-attachment (when an EC is observed for the first time) and re-wiring (when an EC re-wires).

Figure 4 shows the fractions of the providers of malicious ECs that belong to the most preferred providers of non-reported ECs, for the cases of first attachment and re-wiring across time. We observe that for the case of first attachment, all or almost all of the providers they link with are also among the most preferred providers of non-reported ECs. In contrast, for the case of re-wiring, only a percentage of the providers that malicious ECs link with are also among the most preferred providers of non-reported ECs. We observe that for the years 2009–2010, approximately 30–50% of the providers that malicious ECs link with are also among the most preferred providers of non-reported ECs.

5 Recommendations and Future Work

Although we have focused on *characterization* of re-wiring activity, we believe that routing behavior associated with malicious networks has properties that can lead to both better protection against Internet attacks and more efficient detection of the networks that host the attackers. Specifically, because it is independent of any particular attack, routing behavior may serve as invariant behavior that can help identify networks that perpetrate a wider variety of attacks. Routing data is also publicly available and can complement other types of detection mechanisms. In the remainder of this section, first we discuss various lessons from this characterization study that may ultimately inform detection methods and second we discuss future work.

We first note that networks that consistently host malicious activity have re-wiring behaviors that are distinct from other networks. This may point to possibilities for stemming the tide of attack traffic in the future. In the past, we have seen dramatic de-peering events of a single AS (e.g., Atrivo/InterCage), which have resulted in a precipitous drop in spam traffic, which returned at a later date, likely as the spammers re-established upstream connectivity with other upstream ASes. Our analysis shows that the stub ASes that tend to originate a significant amount of attack traffic tend to re-wire with upstream ASes that are not “preferred” upstream ASes. In the future, more effective AS reputation systems might incorporate information not only about the AS itself that originates the traffic, but also the *upstream ASes to which the network connects*. Given the aggressive re-wiring in which malicious ASes participate, another possible way forward might be to encourage some type of registration or verification process by which an AS (or its upstream) is vetted as a reputable provider.

In future work, we plan to evaluate (1) the possibility to classify malicious ASes using re-wiring activity features we observed in this study and (2) the efficacy of using routing information as input to an AS reputation system. We also plan to investigate the re-wiring activity of malicious ASes by class (e.g., small transit providers, large transit providers and content providers) and also by the type of attack they appear to be engaged with (e.g., some malicious ASes may appear mostly to send spam during a specific period of time whereas other ASes may appear to be hosting command-and-control infrastructure). It remains to be seen whether a blacklisting system that is based on re-wiring behavior would be sufficiently different than one that uses observations of the attacks themselves as input.

6 Conclusion

Although limited empirical studies have suggested that attackers exploit BGP routing to help cloak their attacks, there has been no detailed longitudinal study of how malicious networks may interconnect differently from other ASes. In this paper, we have analyzed more than ten years of BGP data in conjunction with reports of malicious ASes from Hostexploit and found that ASes that are known to host malicious traffic consistently exhibit different re-wiring behavior than other ASes. We believe that our findings may ultimately serve as useful features for other reputation or attack detection mechanisms. The fact that routing dynamics are a property of the network hosting the attack, rather

than of any specific attack, may ultimately prove advantageous in this regard. In particular, using BGP routing data as an input to such an AS-based reputation system is a promising area for future work.

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